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SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 40%

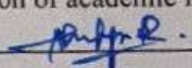
Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I R. Akshaya (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

SHORT ANSWERS TEST

1. Define cloud computing and explain any four characteristics that makes it different from traditional on-premise computing.

Ans:- Cloud computing is the delivery of computing services such as servers, storage, databases, networking, software and analytics over the Internet on pay-as-you-use basis.

Characteristics:

- * On-demand self service - Users can access computing resources whenever needed without human interaction from the provider.
- * Broad network access - Services are available over the internet and can be accessed from devices like laptops, mobiles & tablets.
- * Resource pooling - Computing resources are shared among multiple users using virtualization improving efficiency.
- * Rapid Elasticity - Resources increased or decreased quickly based on demand.

Difference from traditional computing:

Traditional on-premise computing requires organizations to purchase, install, and maintain their own hardware and software, whereas cloud computing provides scalable resources over the Internet with minimal infrastructure management.

2. Trace Evolution of cloud computing from Hardware Evolution to internet software Evolution and show how these changes Enabled modern cloud platforms.

Ans: The Evolution of cloud computing took place in following steps:

1. Hardware Evolution: Large mainframe computers were replaced by personal computers and powerful servers, making computing more affordable.

2. Network Evolution: LANs and the Internet enabled computers to communicate and share resources across different locations.

3. Virtualization: A single physical server could run multiple virtual machines, improving hardware utilization and reducing costs.

4. Internet Software Evolution: Web applications and web services allowed software to be delivered through browsers instead of local installation.

5. Modern cloud Platforms: These technologies combined to create cloud platforms such as AWS, Microsoft Azure, and Google Cloud, offering scalable, reliable and on-demand services.

Example: A company can deploy its website on AWS without purchasing physical servers.

3. Explain the role of server virtualization in cloud computing.

Ans: Server Virtualization:

It is the process of creating multiple virtual servers (Virtual Machines) on a single physical server using a hypervisor. It improves hardware utilization,

reduces cost, save Energy, and simplifies resource management.

Difference :

VIRTUALIZATION	CLOUD COMPUTING.
Technology that creates virtual machines.	Service model that delivers computing resources over the Internet.
Focuses on Efficient use of hardware.	Focuses on providing Scalable Services to users
Implemented using hypervisors like VMware or Hyper V.	Provides IaaS, PaaS, SaaS, other cloud services.

Q1. Role of Data centers:-

Ans:- A data center is a facility containing servers, storage devices, networking equipment and power supplies that host cloud services.

Roles:-

- * stores and process customer data securely.
- * Host cloud applications and websites.
- * Ensure high availability through backup and disaster recovery.
- * Provide scalability by adding servers when demand increases.
- * Maintains security.

5. Differentiate IaaS, PaaS, SaaS, XaaS.

Service Model	Description	Example
IaaS (Infrastructure as a Service)	Provides virtual servers, storage & networking. Users manage OS and applications	Amazon EC2
PaaS (Platform as a Service)	Provides a platform for developing & deploying application without managing infrastructure	Google App Engine.
SaaS (Software as a Service)	Provides ready-to-use software over internet	Microsoft 365
XaaS (Anything as a Service)	Offers any IT service through cloud & databases	Microsoft Azure AI services.